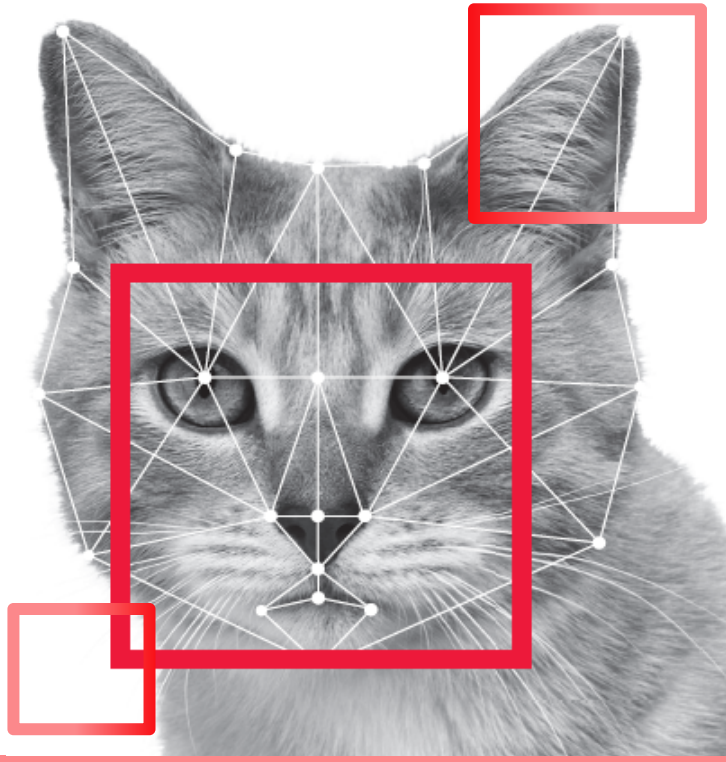
COMPUTER VISION 컴퓨터 비전

기본 개념부터 최신 모바일 응용 예까지



2장. 영상 처리

각 절에서 다루는 내용

1. 디지털 영상이란?
2. 히스토그램
3. 이진 영상
4. 영상 처리의 세 가지 기본 연산
5. 다해상도
6. 모폴로지
7. 컬러

PREVIEW

- Eye and Camera
- Image formation
- Camera model
  - Pinhole camera model
  - Perspective projections
  - Focal length and filed of view

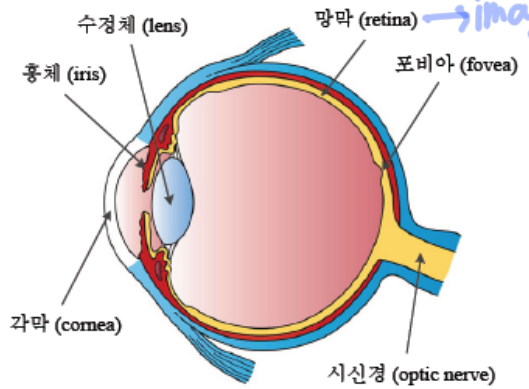
2.1 디지털 영상이란?

2.1.1 디지털 영상의 태동

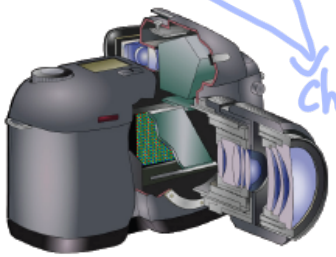
2.1.2 획득과 표현

2.1.2 획득과 표현

- 사람의 눈과 카메라
  - 수정체는 렌즈, 망막은 CCD 센서 (필름)에 해당

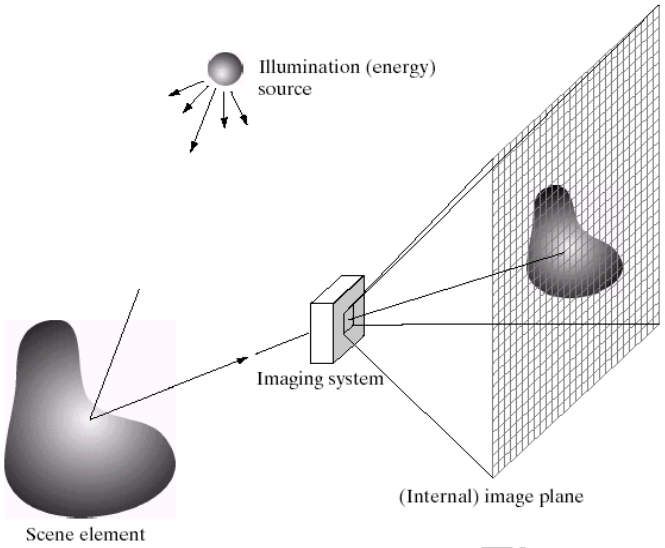


(a) 사람의 눈 구조

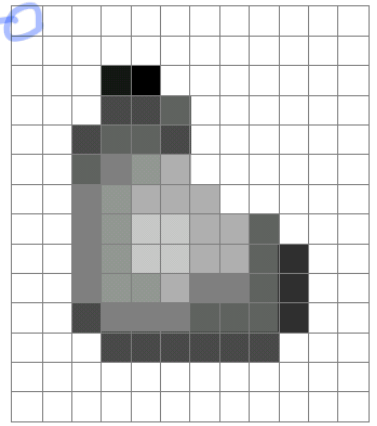


(b) 카메라의 구조

그림 2-3 사람의 눈과 카메라의 구조



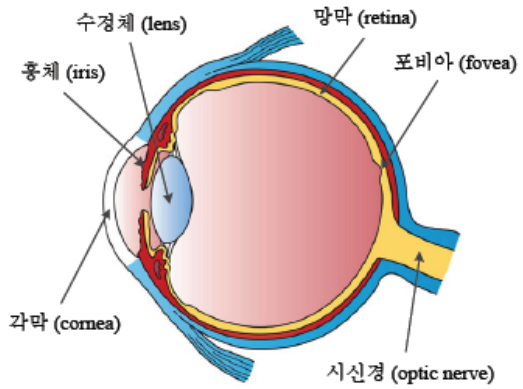
Film



Digital Camera

2.1.2 획득과 표현

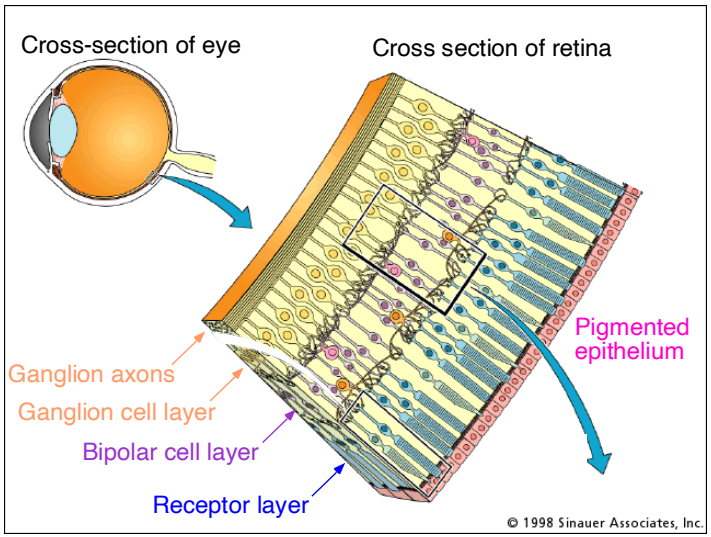
- The Eye



(a) 사람의 눈 구조

그림 2-3 사람의 눈과 카메라의 구조

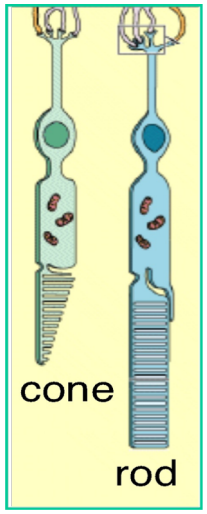
망막(Retina)



빛 ↓ Rods 활성화↑

R,G,B 중에 G에 가장 민감하게 반응

Light ↓↓↓



Cones

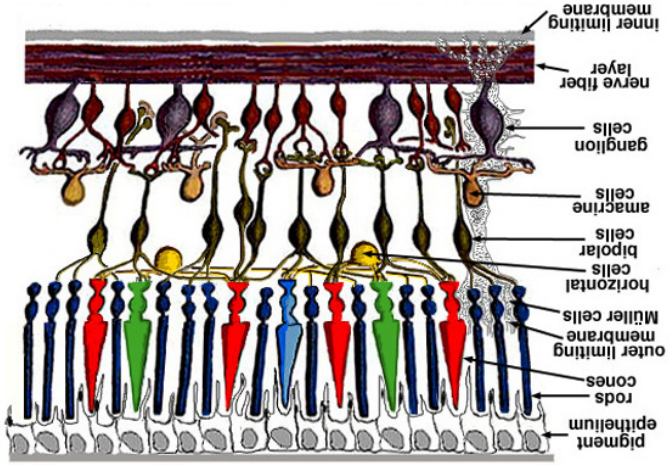
cone-shaped less sensitive
operate in high light
color vision

Color 인식 시신경

Rods

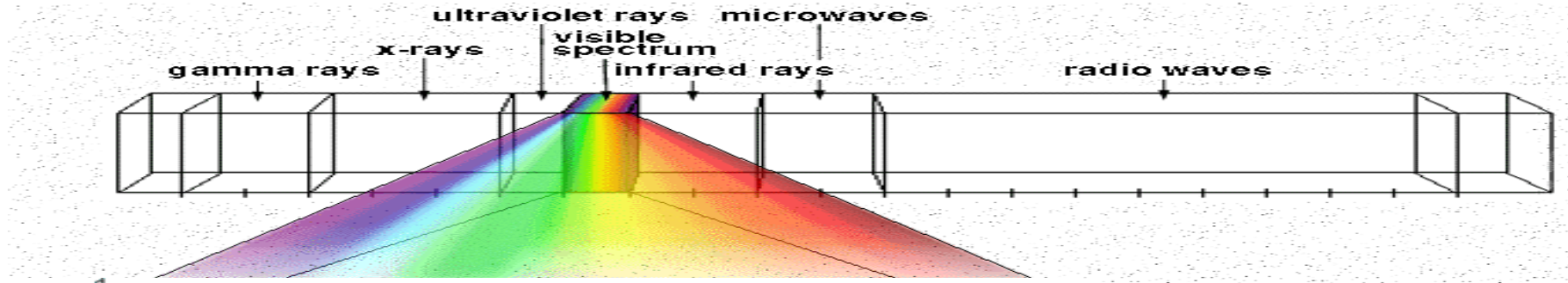
rod-shaped highly sensitive
operate at night gray-scale vision

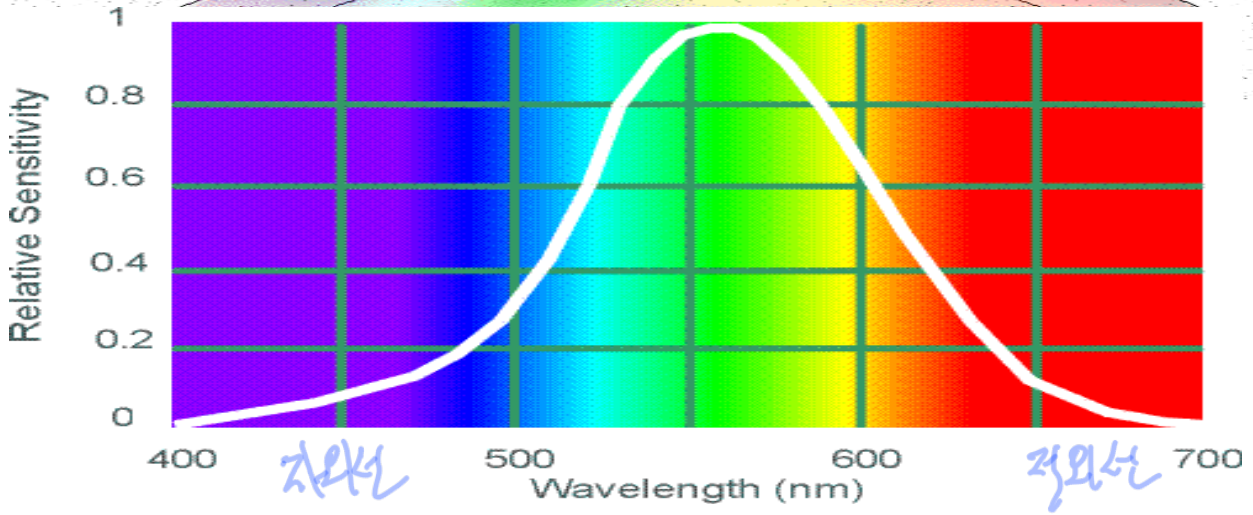
어둡고 빛은 것 인식하는 시신경



2.1.2 획득과 표현

- Electromagnetic spectrum





특정 주파수의 빛만 볼 수 있다

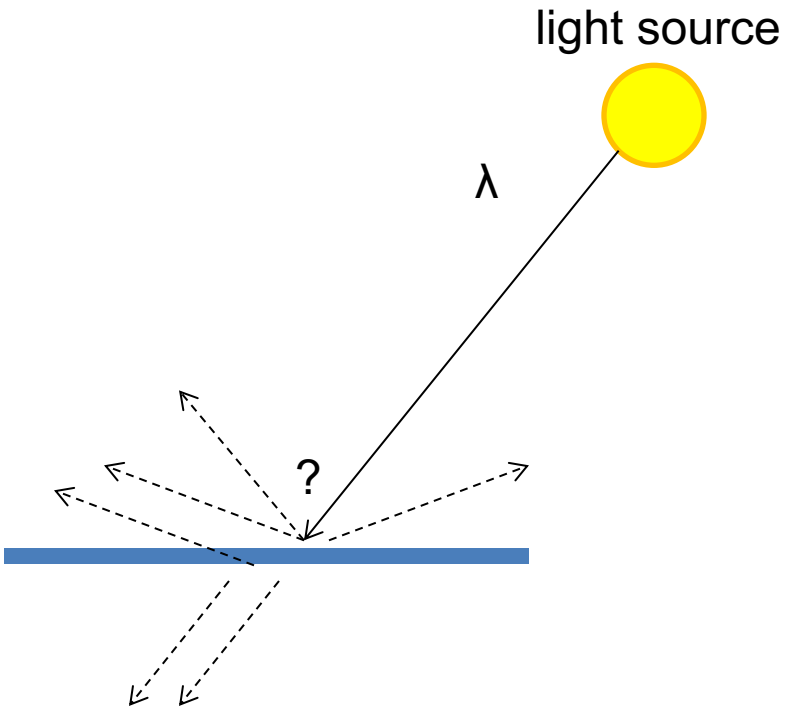
└가시광선

Human Luminance Sensitivity Function

A photon's life choices

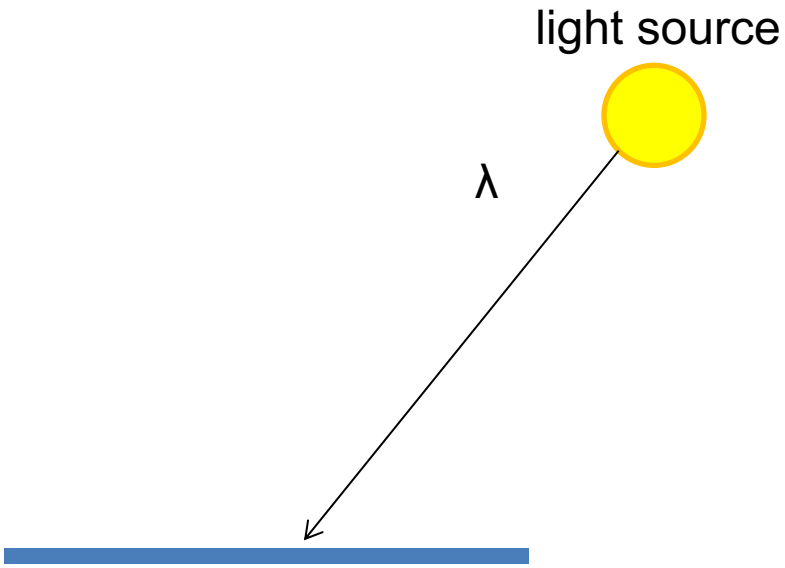
- Absorption
- Diffusion
- Reflection
- Transparency
- Refraction
- Fluorescence
- Subsurface scattering
- Phosphorescence
- Interreflection

반사 흡수 통과



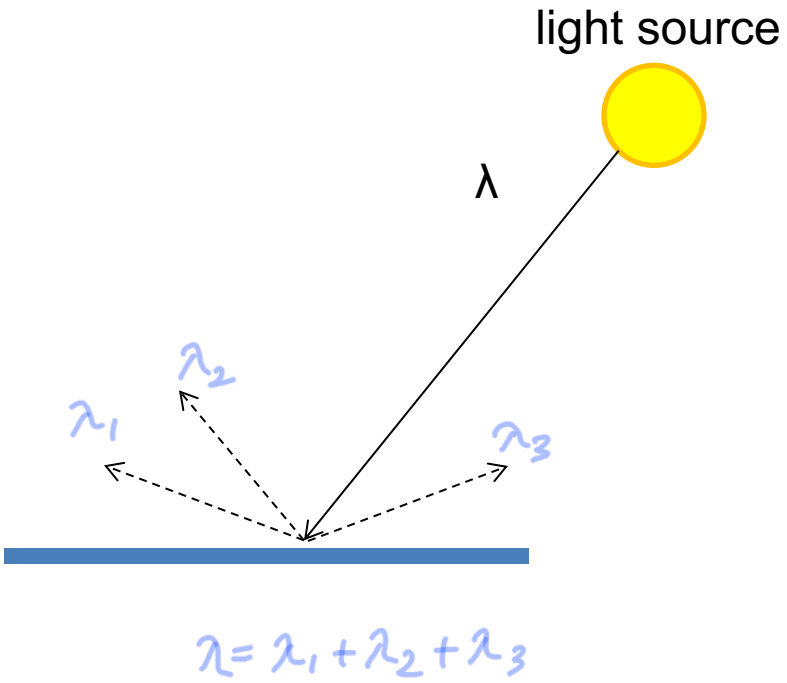
A photon's life choices

- **Absorption** 흡수
- Diffusion
- Reflection
- Transparency
- Refraction
- Fluorescence
- Subsurface scattering
- Phosphorescence
- Interreflection



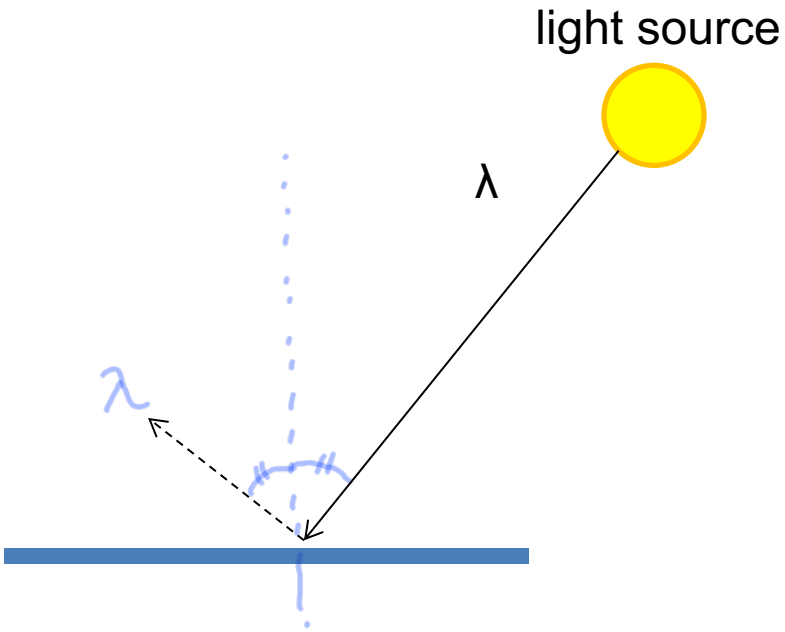
A photon's life choices

- Absorption
- **Diffuse Reflection**
- Reflection
- Transparency
- Refraction
- Fluorescence
- Subsurface scattering
- Phosphorescence
- Interreflection



A photon's life choices

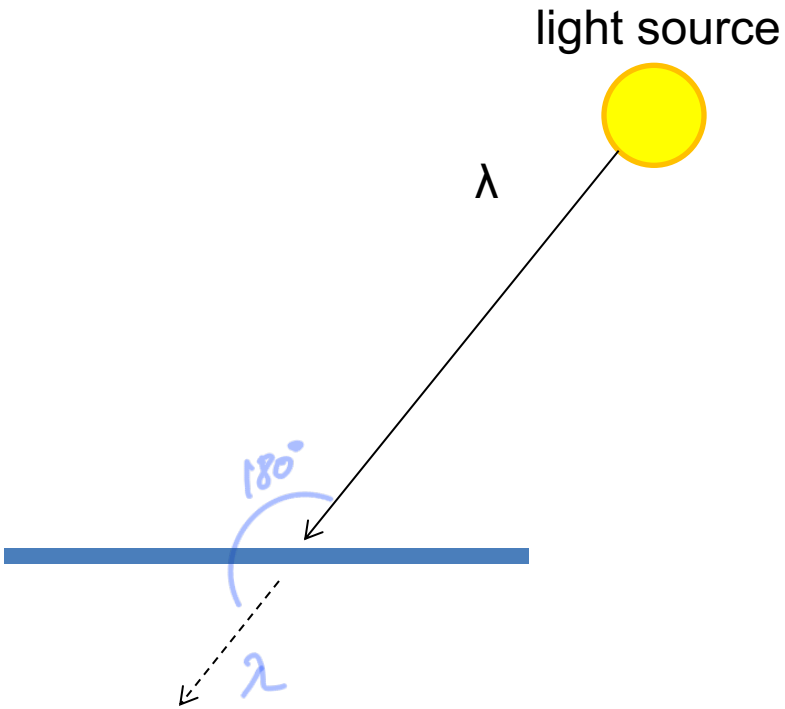
- Absorption
- Diffusion
- **Specular Reflection**
- Transparency
- Refraction
- Fluorescence
- Subsurface scattering
- Phosphorescence
- Interreflection



카메라의 색상은 여기에 해당

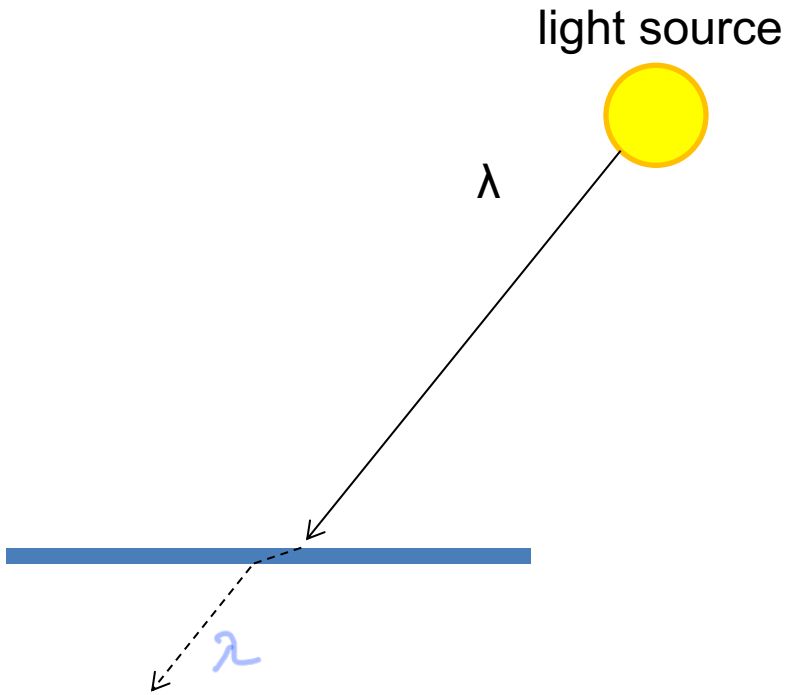
A photon's life choices

- Absorption
- Diffusion
- Reflection
- **Transparency**
- Refraction
- Fluorescence
- Subsurface scattering
- Phosphorescence
- Interreflection



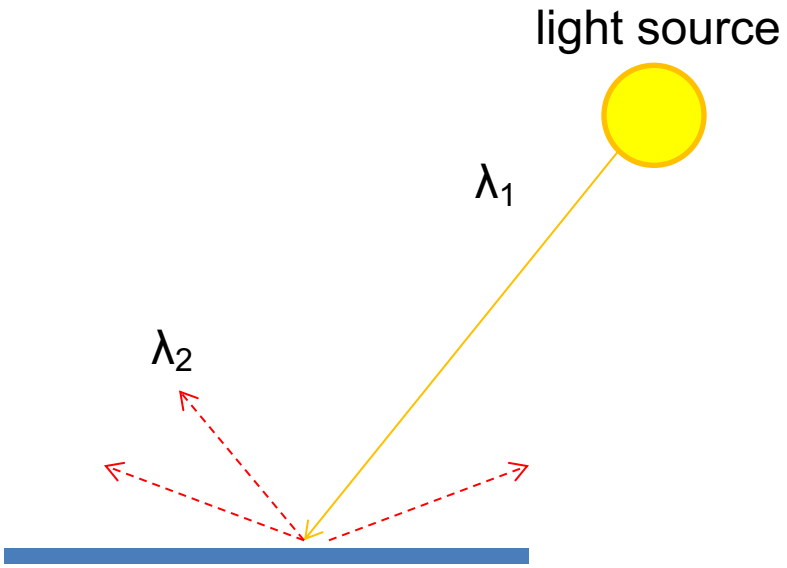
A photon's life choices

- Absorption
- Diffusion
- Reflection
- Transparency
- **Refraction**
- Fluorescence
- Subsurface scattering
- Phosphorescence
- Interreflection



A photon's life choices

- Absorption
- Diffusion
- Reflection
- Transparency
- Refraction
- **Fluorescence**
- Subsurface scattering
- Phosphorescence
- Interreflection

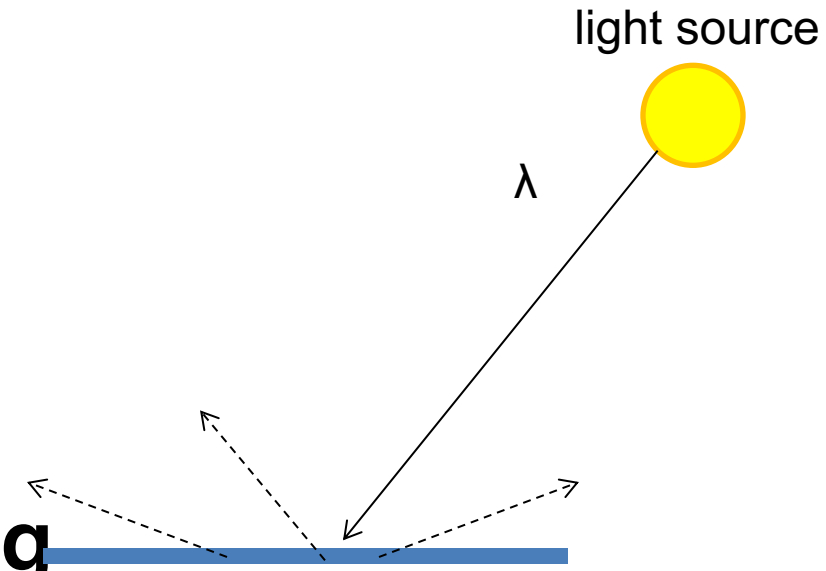


$\lambda_2 < \lambda_1$

$\lambda_1 - \lambda_2 \Rightarrow$ 흡수

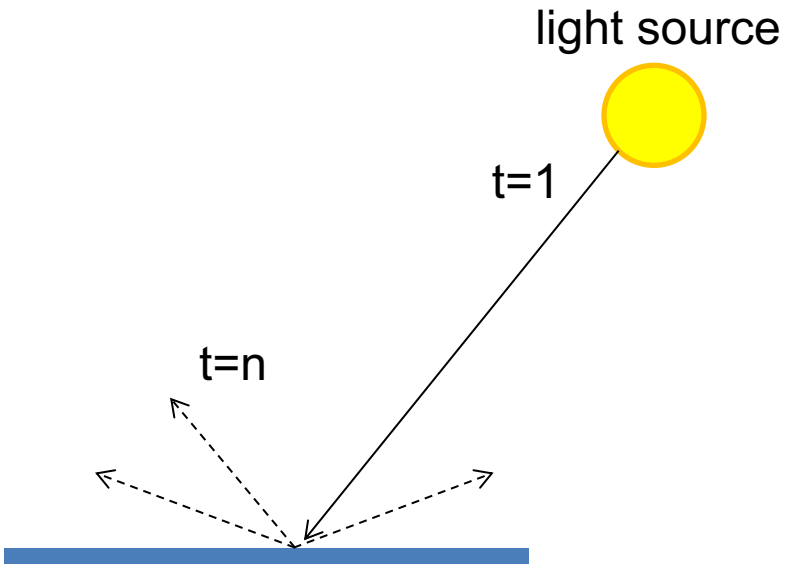
A photon's life choices

- Absorption
- Diffusion
- Reflection
- Transparency
- Refraction
- Fluorescence
- **Subsurface scattering**
- Phosphorescence 난반사
- Interreflection



A photon's life choices

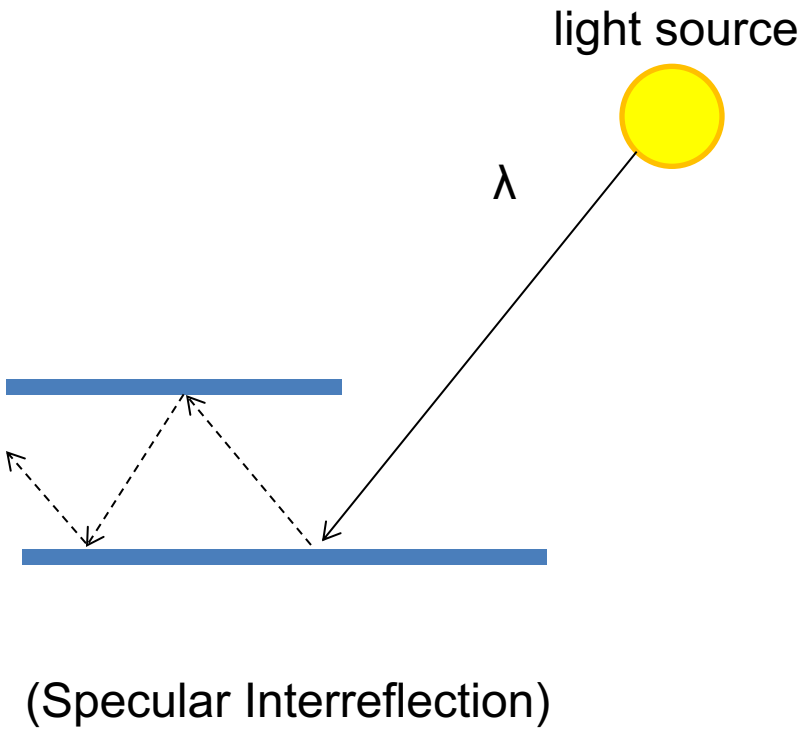
- Absorption
- Diffusion
- Reflection
- Transparency
- Refraction
- Fluorescence
- Subsurface scattering
- **Phosphorescence**
- Interreflection



일정 시간 흡수 뒤 방출

A photon's life choices

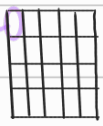
- Absorption
- Diffusion
- Reflection
- Transparency
- Refraction
- Fluorescence
- Subsurface scattering
- Phosphorescence
- **Interreflection**



광통신

필름 → 아날로그
센서 → 디지털

0 ~ 255
black white



256 = 2^8 : 8-bit

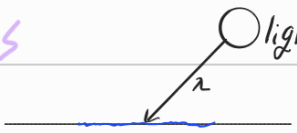
Cones : Color 인식 시신경
Rods : 어둡고 밝음 인식 시신경

화소 (pixel)
Picture element

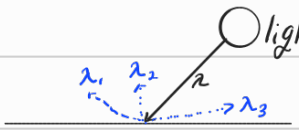
밝다 → Rods 활성화↑
R, G, B 중에 G에 가장 민감하게 반응
Green
배경을 초록으로 하는 이유

photon's life choices

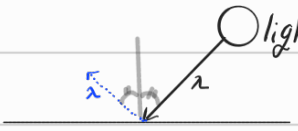
: absorption 흡수



: Diffuse Reflection 확산 반사

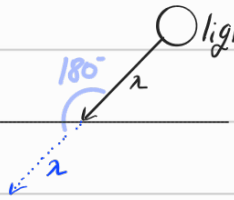


: Specular Reflection 정반사

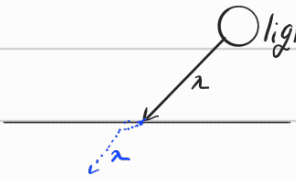


빛을 받은 물체는 투과하지도
흡수하지도 않고 받은 양만큼
반사각과 대칭으로 그대로 반사
⇒ 컴퓨터 비전 핵심 논리

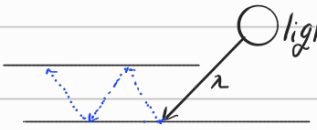
: transparency 투과
그대로 통과



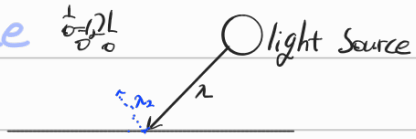
※ 컴퓨터 비전의 기본
: Refraction 굴절



: Interreflection 상호반사
여러 번 반사



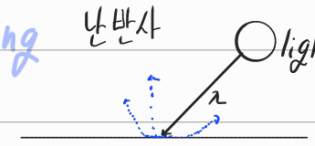
Fluorescence 형광



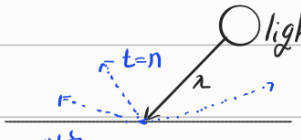
$\lambda > \lambda_2$

$\lambda - \lambda_2 =$ 흡수

: Subsurface scattering 난반사



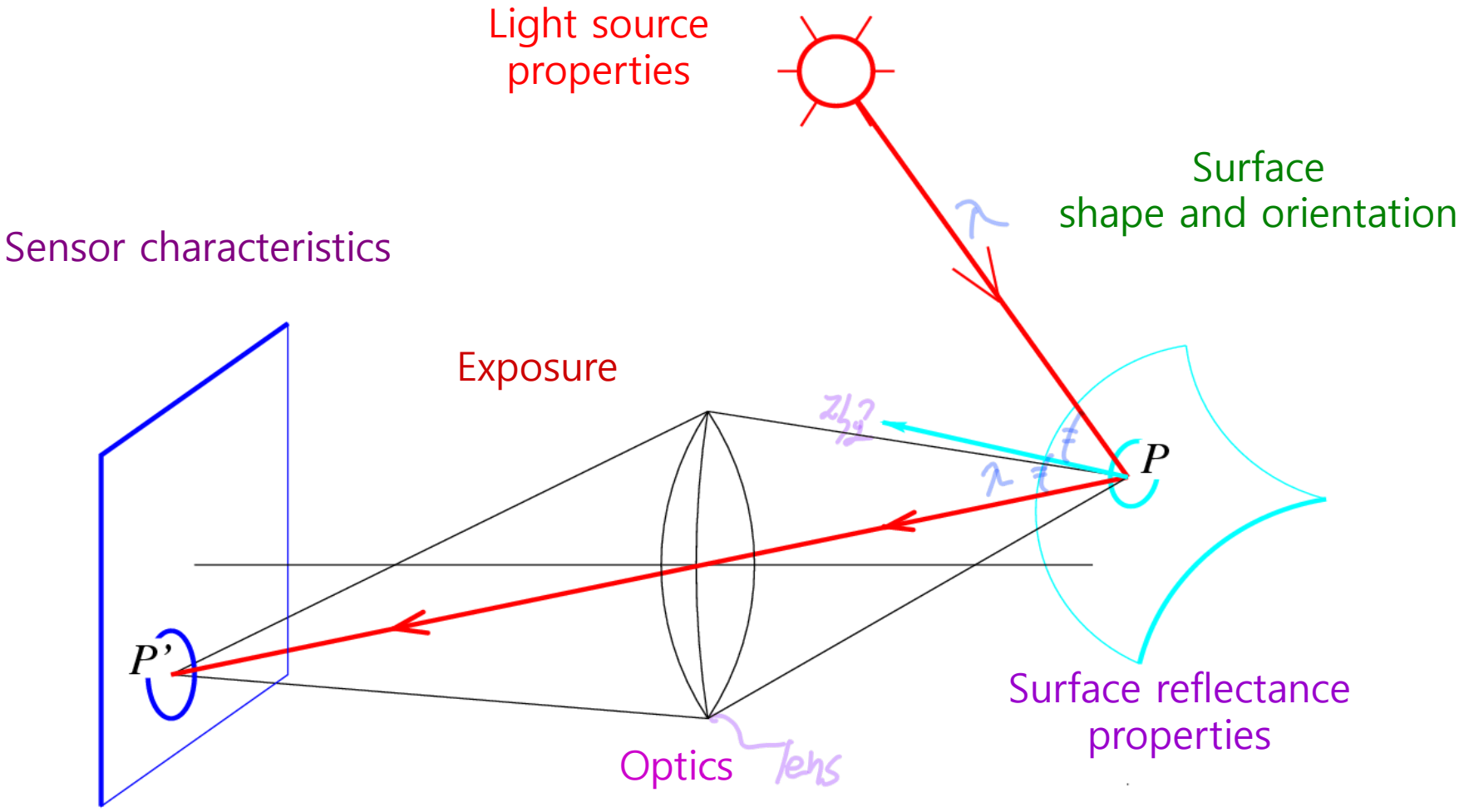
: phosphorescence 인광



일정 시간 흐른 뒤 방출

2.1.2 획득과 표현

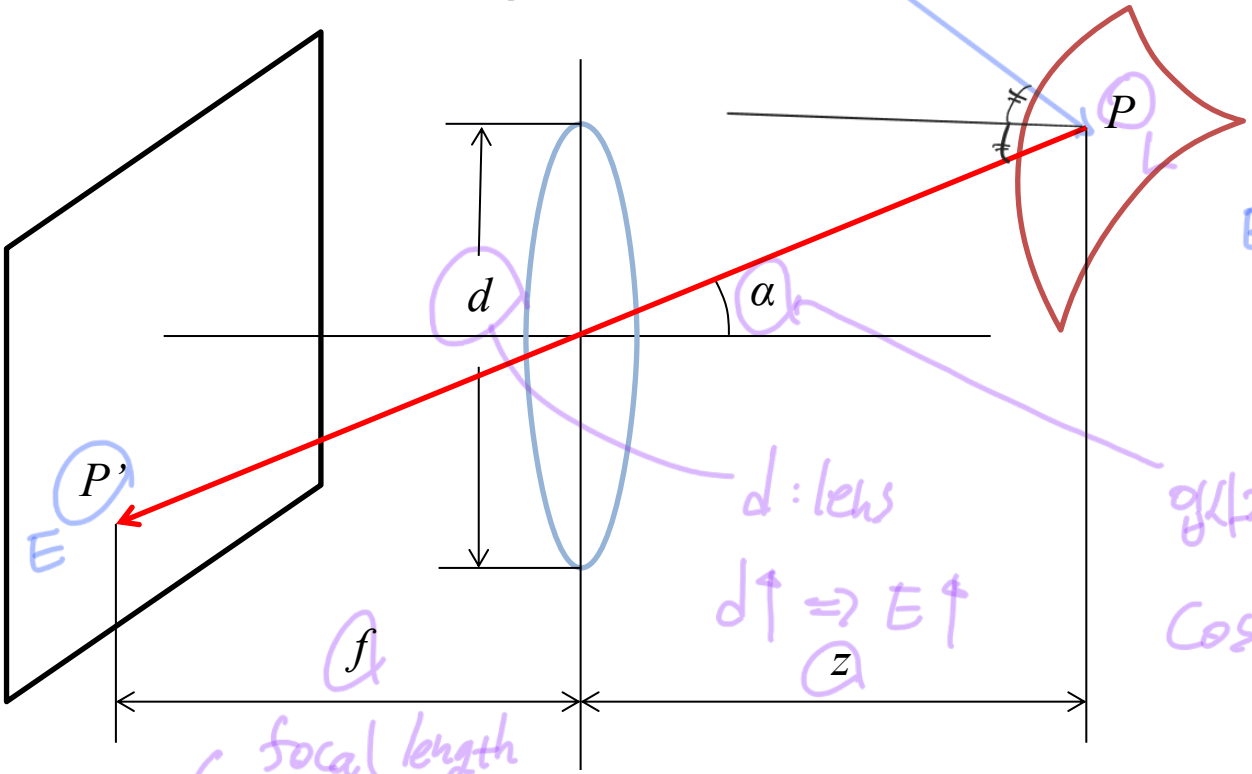
- Image formation



Slide by L. Fei-Fei

Fundamental radiometric relation

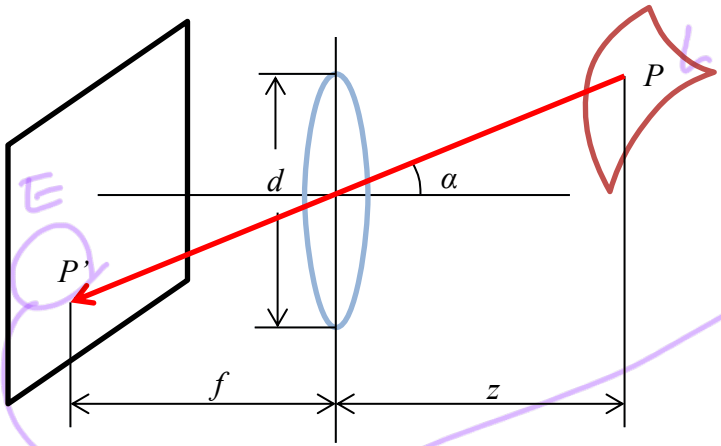
- *L*: Radiance emitted from *P* toward *P'*
- *E*: Irradiance falling on *P'* from the lens



What is the relationship between *E* and *L*?

Szeliski 2.2.3

Fundamental radiometric relation

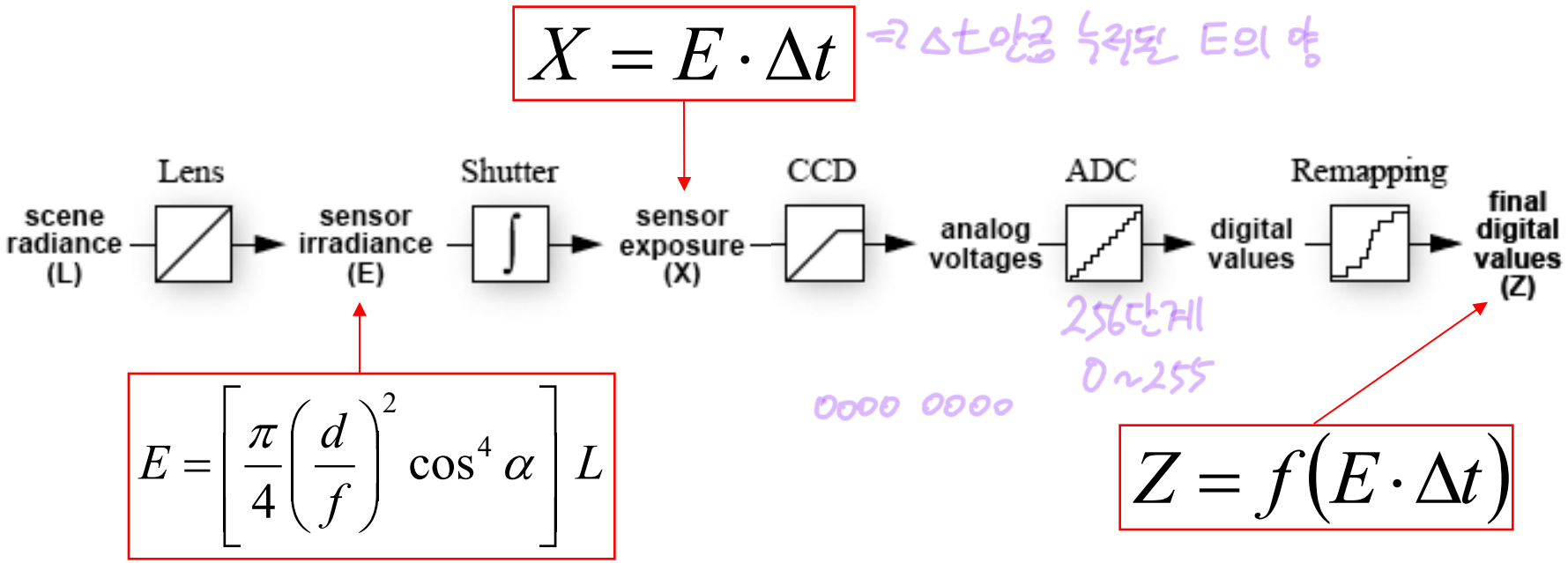


$$E = \left[\frac{\pi}{4} \left(\frac{d}{f} \right)^2 \cos^4 \alpha \right] L$$

- Image irradiance is linearly related to scene radiance
- Irradiance is proportional to the area of the lens and inversely proportional to the squared distance between the lens and the image plane
- The irradiance falls off as the angle between the viewing ray and the optical axis increases

Szeliski 2.2.3

From light rays to pixel values



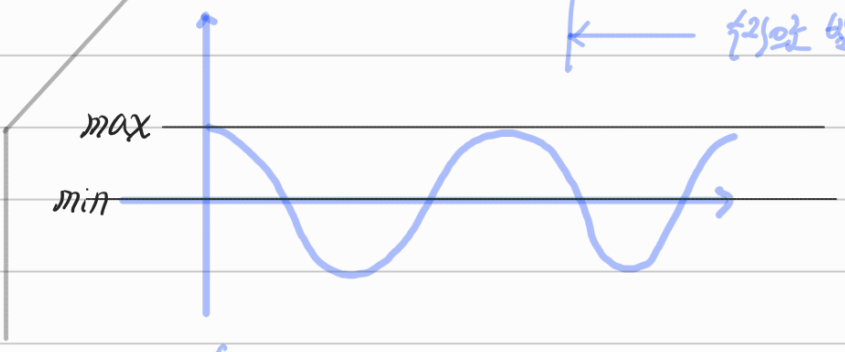
- Camera response function: the mapping f from irradiance to pixel values
  - Useful if we want to estimate material properties
  - Enables us to create high dynamic range images
  - For more info: P. E. Debevec and J. Malik, *Recovering High Dynamic Range Radiance Maps from Photographs*, *SIGGRAPH 97*

lens 크기 ↑ ⇒ 들어오는 빛의 양 ↑
d E

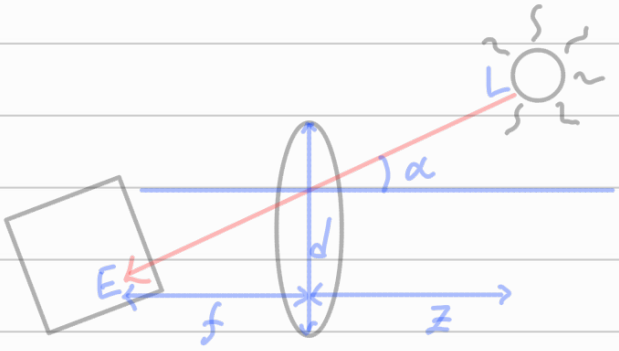
(lens 크기 ∞ ⇒ E = L)

빛이 들어오는 입사각 ↓ ⇒ E ↑
α

(max 값 ⇒ cos α = 1 (0°)
← 수직으로 빛이 들어옴



(min 값 ⇒ cos α = 0 (90°)
‖ 평행하게 빛이 들어옴



focal length
∴ f ↓ ⇒ E ↑

$$E = \left[\frac{\pi}{4}\left(\frac{d}{f}\right)^2 \cos^4\alpha\right] L$$ sensor irradiance

↑ lens

scene radiance
(L) 빛의 양

↓ shutter

$X = E \cdot \Delta t$
sensor exposure

↑ CCD

analog Voltage

↑ ADC

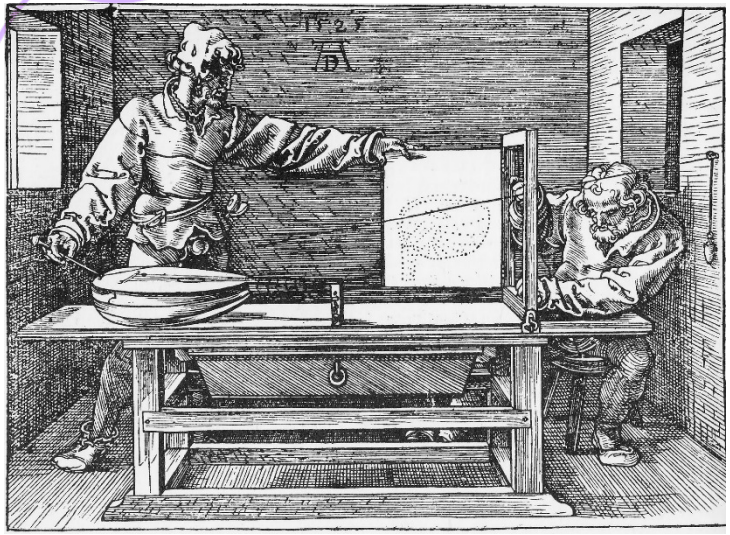
digital value

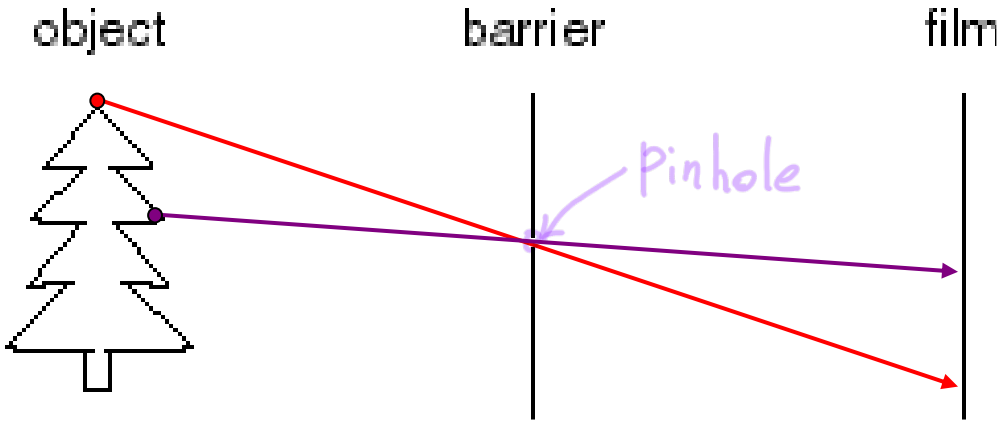
↓ remapping

$Z = f(E \cdot \Delta t)$
final digital values

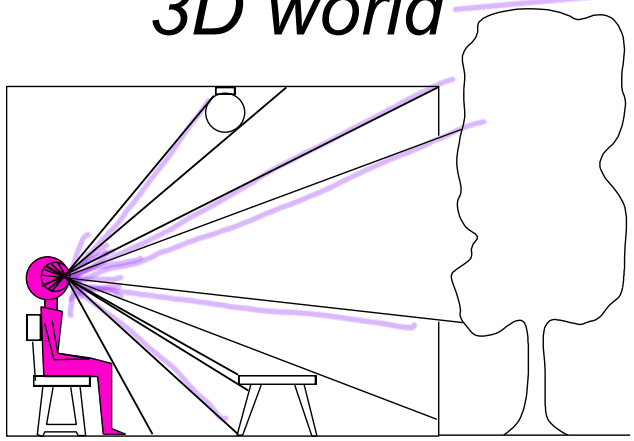
2.1.2 획득과 표현

- Perspective projection



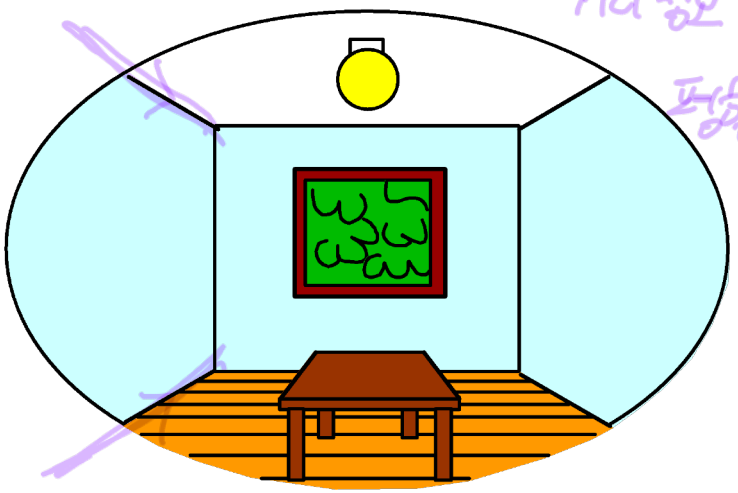


3D world



Point of observation

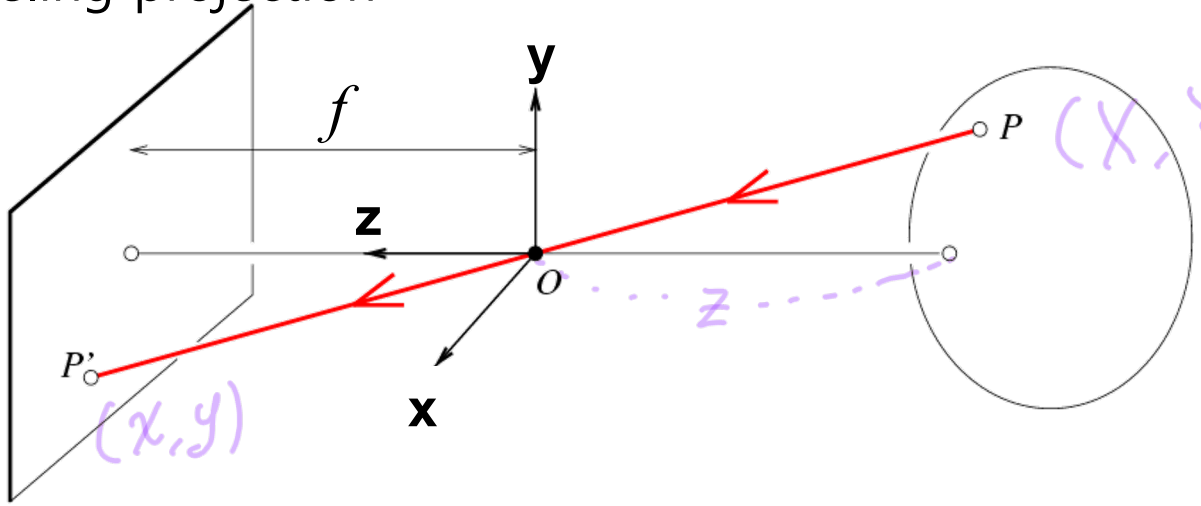
2D image



2.1.2 획득과 표현

기하학적 관계 (3D와 2D)

- Modeling projection



(X, Y, Z)

z

(u, v) (x, y)

  - 3차원 공간상에서의 P는 2차원 영상 평명(image plane)에서 P′으로 투영됨
  - P와 P′의 상관관계를 나타낼 수 있는 방법 필요
    - Optical center (**O**) 가 3차원 공간상의 원점
    - Projection equations

$f : z = x : X$

$f : z = y : Y$

$$(x, y, z) \rightarrow (f\frac{x}{z}, f\frac{y}{z})$$

수학 : 물체 더 잘 표현 | 공학 : 물체 더 잘 묘현 (더 크게 보임)

2.1.2 획득과 표현



- Perspective projection matrix

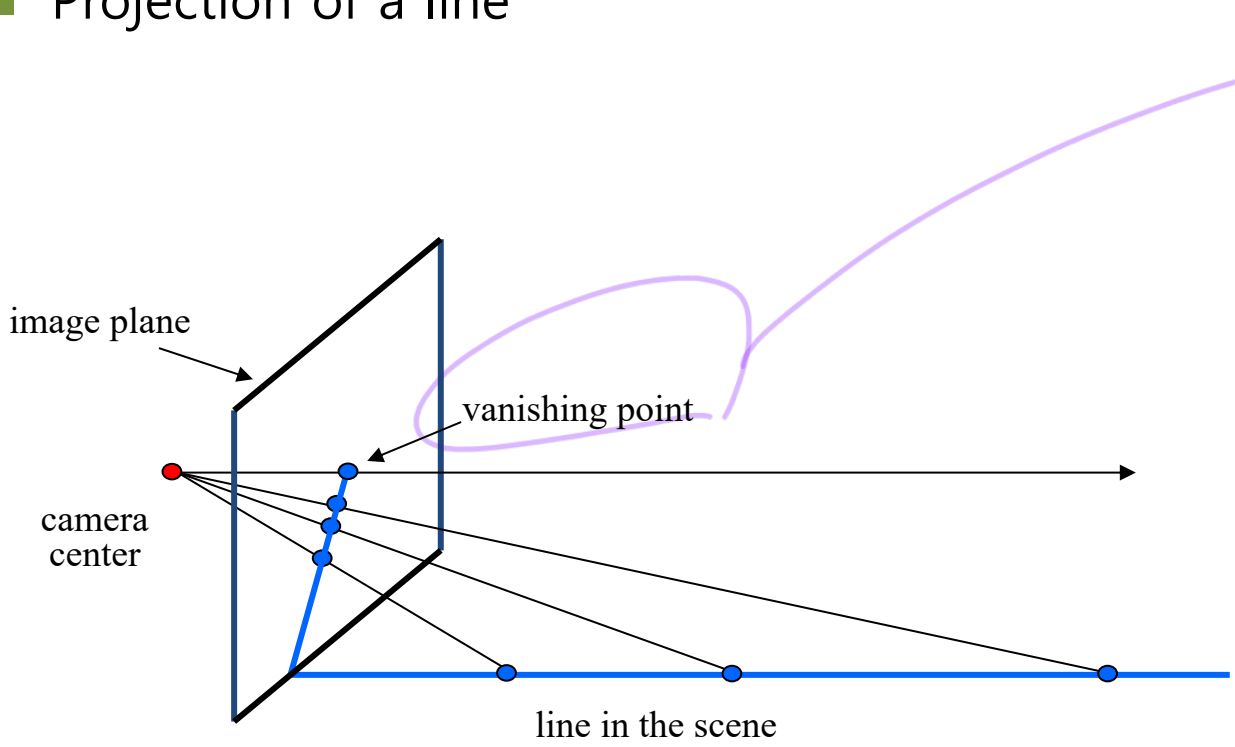
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/f & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z/f \end{bmatrix} \Rightarrow \left(f\frac{x}{z}, f\frac{y}{z}\right)$$

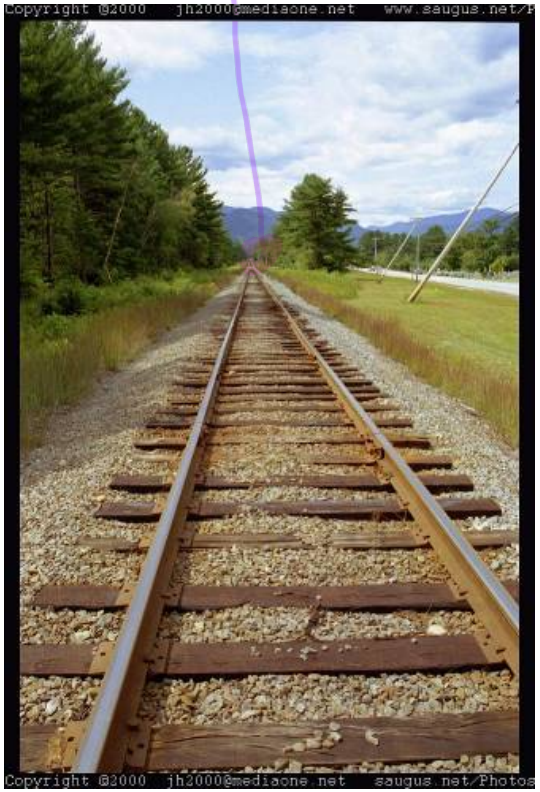
divide by the third co ordinate

$$\begin{pmatrix} \text{2D point (3x1)} \end{pmatrix} = \begin{pmatrix} \text{Camera to pixel coord. trans. matrix (3x3)} \end{pmatrix} \begin{pmatrix} \text{Perspective projection matrix (3x4)} \end{pmatrix} \begin{pmatrix} \text{World to camera coord. trans. matrix (4x4)} \end{pmatrix} \begin{pmatrix} \text{3D point (4x1)} \end{pmatrix}$$

2.1.2 획득과 표현

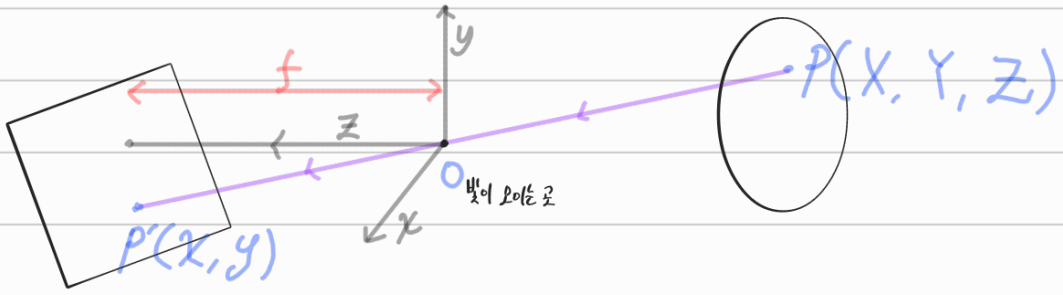
- Projection of a line





perspective projection 원근 투영

3D → 2D : depth, 거리 정보 z 소실



$$(X, Y, Z) \longrightarrow \left(f\frac{X}{Z}, f\frac{Y}{Z}\right)$$

z : 렌즈 → 피사체까지 거리 $z\downarrow$ ⇒ 물체 더 잘 표현 (더 크게 보임)

f : 렌즈 → 센서(이미지 평면)까지 거리 $f\uparrow$ ⇒ 물체 더 잘 표현 (더 크게 보임)
but 타, 밝기는 줄어든다
더 많은 픽셀 처리 → 해상도↑

$f : z = x : X$
$\Rightarrow x = f \cdot \frac{X}{z}$

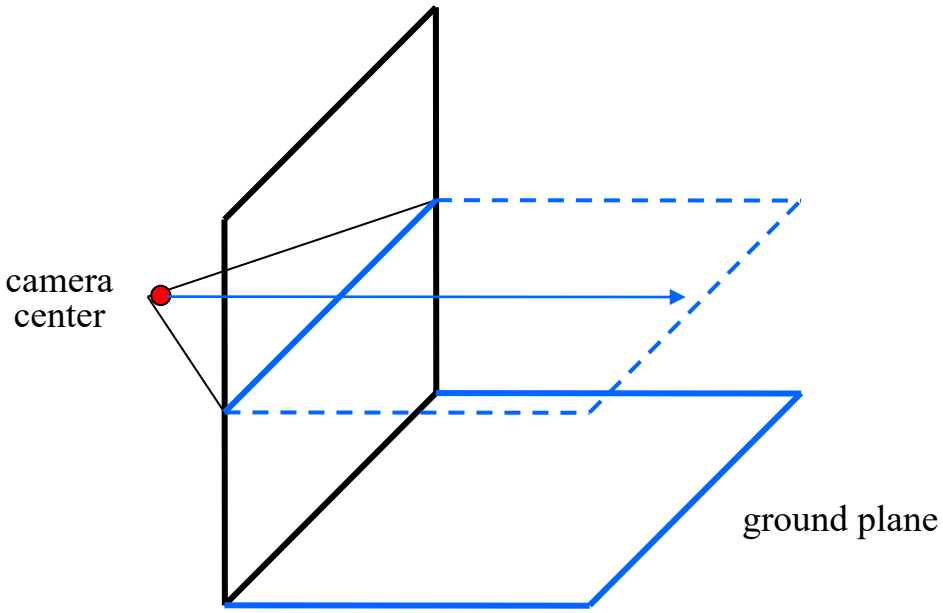
$f : z = y : Y$
$\Rightarrow y = f \cdot \frac{Y}{z}$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/f & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} X \\ Y \\ Z/f \end{bmatrix} \Rightarrow \left(f\cdot\frac{X}{Z}, f\cdot\frac{Y}{Z}\right)$$



2.1.2 획득과 표현

- The horizon





Take away

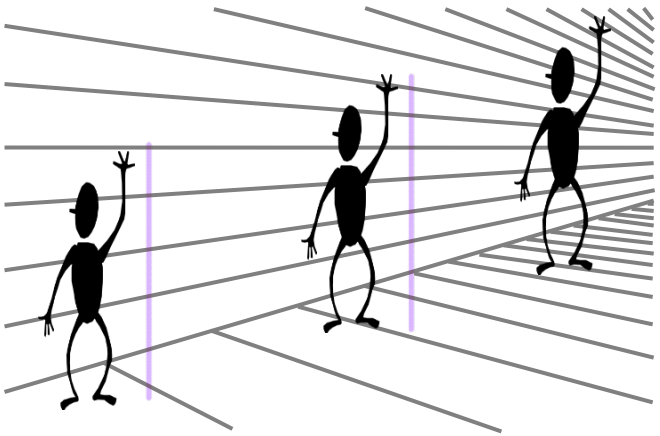
① 사람의 눈/카메라

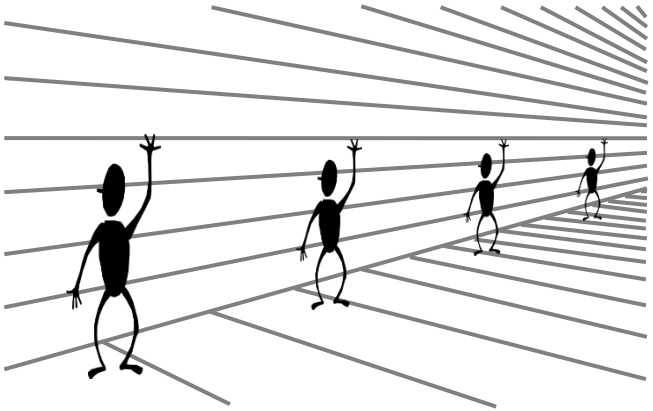
② 빛의

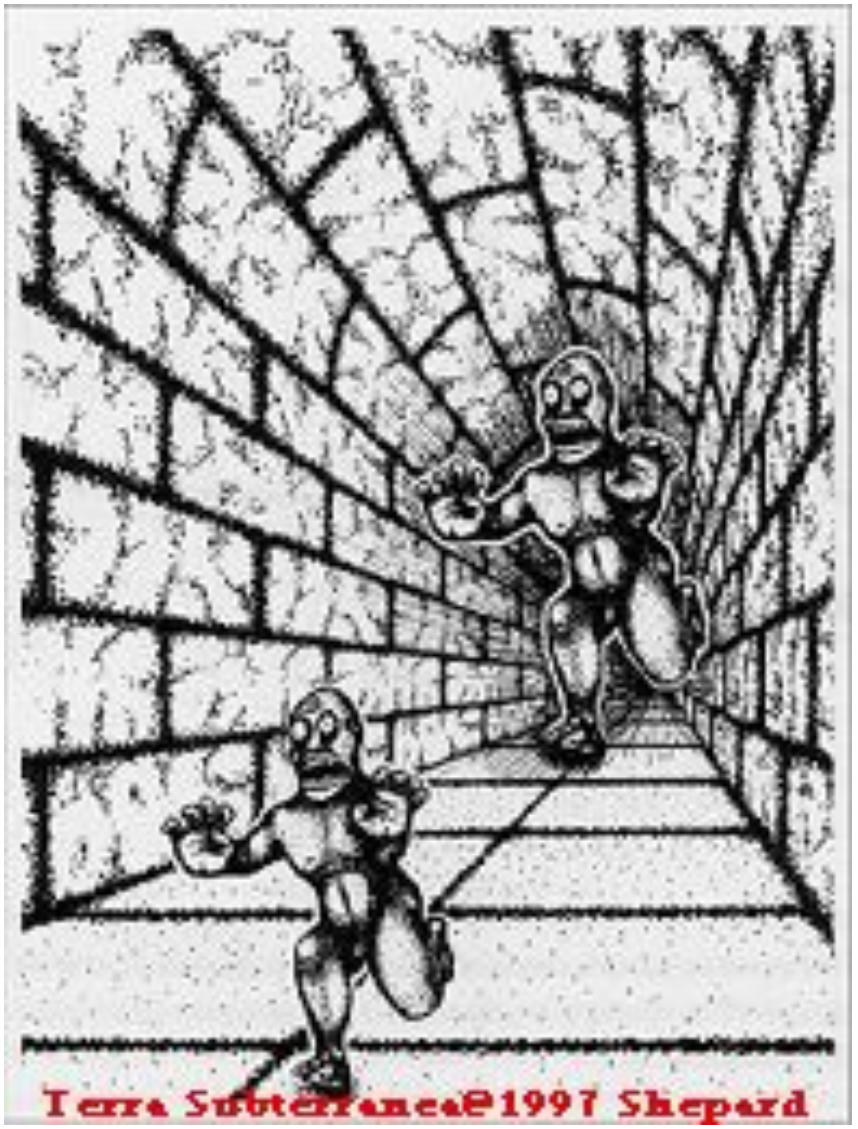
③ 3차원 공간/2차원 영상 간의 관계

2.1.2 획득과 표현

- The perspective cues



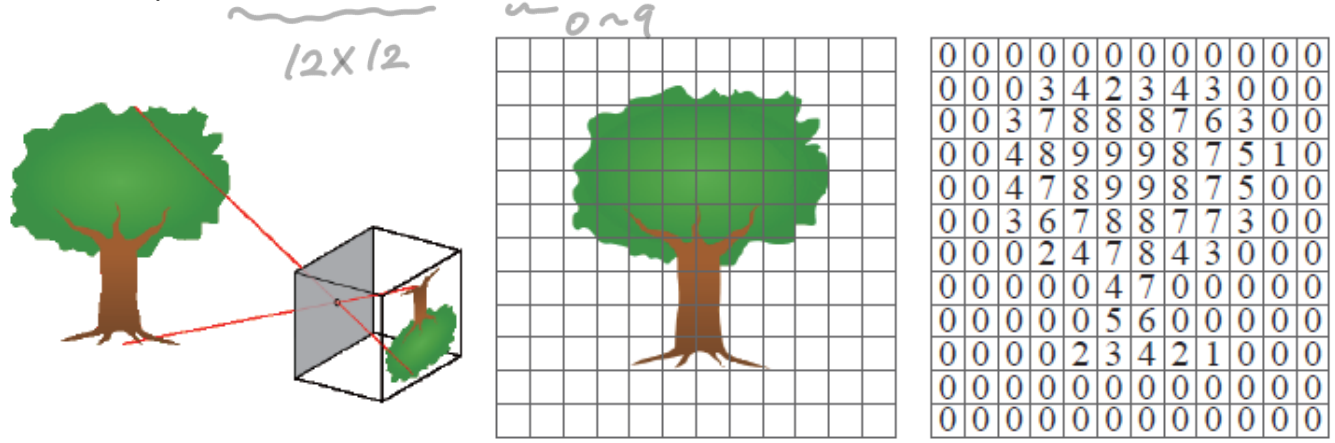




2.1.2 획득과 표현

샘플링과 양자화

- 2차원 영상 공간을 *M***N*으로 샘플링 (*M***N*을 해상도라 부름)
- 명암을 *L* 단계로 양자화 (*L*을 명암 단계라 부름, 즉 명암은 [0,*L*-1] 사이 분포)
- 아래 예) *M*=12, *N*=12, *L*=10인 경우



0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	3	4	2	3	4	3	0	0	0
0	0	3	7	8	8	8	7	6	3	0	0
0	0	4	8	9	9	9	8	7	5	1	0
0	0	4	7	8	9	9	8	7	5	0	0
0	0	3	6	7	8	8	7	7	3	0	0
0	0	0	2	4	7	8	4	3	0	0	0
0	0	0	0	0	4	7	0	0	0	0	0
0	0	0	0	0	5	6	0	0	0	0	0
0	0	0	0	2	3	4	2	1	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0

(a) 핀홀 카메라 모델 (b) 샘플링과 양자화 (c) 디지털 영상

그림 2-4 디지털 영상 획득

2.1.2 획득과 표현

- 영상 좌표계
  - 화소 위치는 **x**=(*j*,*i*) 또는 **x**=(*y*,*x*)로 표기
  - 영상은 *f*(**x**) 또는 *f*(*j*,*i*), 0≤*j*≤*M*-1, 0≤*j*≤*N*-1로 표기

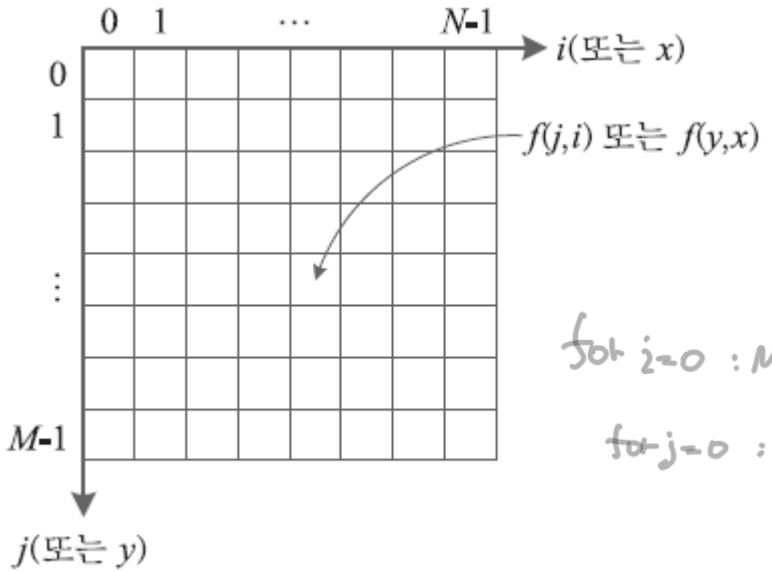


그림 2–5 디지털 영상의 좌표계

(0,0) (0,1) (0,2) … (0,N-1)

for i=0 : M

for j=0 : N

(M-1,0) . . . (M-1,N-1)

- 컬러 영상은 fr(x), fg(x), fb(x)의 세 채널로 구성





평행한 수많은 직선들의 소실점
⇒ 지평선

2차원 영상 공간
→ M * N 으로 샘플링
명암을 L 단계로 양자화 [0, L-1]

ex) M=4, N=4, L=3

	0	1	2	3
0	0	1	2	0
1	1	0	2	0
2	0	2	1	2
3	2	1	1	1

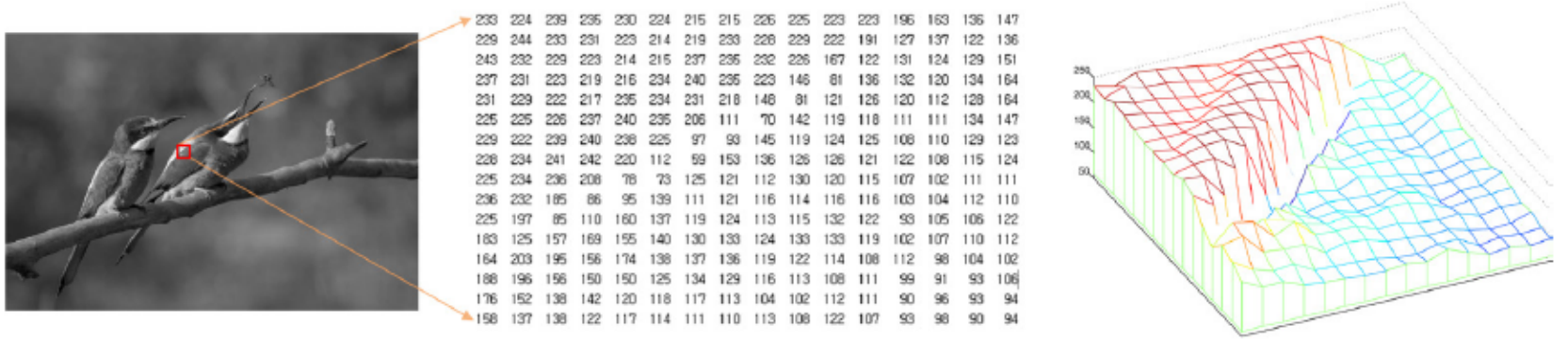
N

M

(0, 0) ··· (0, N-1)
⋮ ⋮
(M-1, 0) ··· (M-1, N-1)

2.1.2 획득과 표현

- 영상 표시 방법



(a) 영상 (b) 숫자 배열 (c) 지형

그림 2-6 디지털 영상 표시 방법